

1. We introduce answer-conditioned CoT distillation for VLMs, where a frontier model generates justified reasoning given the correct answer, and a small model is fine-tuned on these reasoning-augmented examples.
2. We validate the method across four industrial domains spanning three image modalities, showing consistent improvement over direct fine-tuning (+0.9 to +8.3 percentage points).
3. We demonstrate that a 3B-parameter model fine-tuned on 24 images can outperform a frontier model on weld defect classification.
4. We show that this approach addresses a real manufacturing need: rapid adaptation to new inspection tasks when only minimal labeled data is available, with end-to-end training completing in under 2 hours on a single GPU.

2 Related Work

We situate our work at the intersection of four research areas: chain-of-thought distillation, knowledge distillation for VLMs, few-shot industrial vision, and self-adapting language models.

2.1 Chain-of-Thought Distillation

Chain-of-thought (CoT) prompting [15] showed that including intermediate reasoning steps improves performance on complex tasks. Hsieh *et al.* [9] extended this to knowledge distillation, demonstrating that training small language models on reasoning-augmented outputs from larger models outperforms both standard fine-tuning and the teacher model itself, while requiring less data. Their work is the closest precursor to our approach. However, it operates entirely in the text domain. We extend this principle to vision-language models, where the teacher generates image-grounded visual reasoning rather than text-only rationales.

More recent work on CoT distillation for smaller models [16] addresses memorization risks in text-only settings. Our approach avoids this issue through answer-conditioning: by providing the correct label to the teacher, we ensure that the generated reasoning is directed toward the correct conclusion and grounded in the specific visual content of each image.

2.2 Knowledge Distillation for Vision-Language Models

Several recent papers address distillation in the VLM setting. VLsI [9] distills internal representations from large VLMs to smaller ones using natural language verbalizers at each layer. This differs from our approach, which distills output-level reasoning rather than intermediate activations. Knowledge distillation from VLMs for long-tail recognition [17] uses logit-based distillation to transfer knowledge about rare classes. Our method uses text-based reasoning transfer, which is more interpretable and does not require access to teacher model logits. Online In-Context Distillation [8] maintains the teacher at inference time for collaborative prediction. We bake the teacher’s knowledge into the student’s weights, eliminating inference-time dependency on the frontier model.

None of these methods use answer-conditioned reasoning generation as the distillation mechanism.